

PONTIFICAL CATHOLIC UNIVERSITY OF PARANÁ

DATA SCIENCE

PROF. JEAN PAUL BARDDAL
PROF. RAYSON LAROCA



Outline

Data science projects involve multiple steps that include data acquisition, data engineering, data cleansing, data analysis, machine learning, and validation. In this project, you must assemble in teams and work on a data science project that targets creating and assessing classification, regression, or clustering machine learning models. During the semester, each team must:

1. **Define a dataset and task:** define a dataset that can be used in either classification, regression, or clustering tasks. Make sure the dataset is publicly available since the link for downloading the dataset must be made available. If the dataset is not retrieved from a traditional webpage (UCI, Kaggle, HuggingFace, etc.), make sure the data is downloaded for posterior delivery and evaluation;
2. **Acquire the dataset:** download the dataset. If your dataset is too big (more than 1 million instances), consider acquiring a data sample so that computational resources do not become prohibitive. If this is the case, make sure the sampling process is adequate. After data engineering and cleansing, the dataset must have at least 30 features and 500 instances;
3. **Data analysis:** conduct an exploratory data analysis. In this step, make sure you have at least a descriptive analysis using statistics that describes the basic characteristics of the dataset (instances, features, class ratio, etc.), 20 univariate analyses, 10 multivariate analyses, and 5 effective data visualizations. All analyses must be given in the question/hypothesis-analysis-discussion format;
4. **Data engineering and data cleansing:** represent your dataset in an adequate manner. Make sure you have tabular data in the format machine learning libraries require. Work on your data so that you have enough features for posterior analysis and machine learning development. Next, work on data skewness, scaling, standardization, missing data, outliers, class imbalance, etc.
5. **Machine learning:** apply machine learning algorithms and evaluate them using an adequate experimental protocol (validation process and evaluation metrics). Consider hyper-parameter tuning in this step. The definition of the entire experimental protocol must be explained and adhere to the dataset's characteristics and context;

6. **Validation:** after obtaining the results of different machine learning algorithms, compare them and draft a conclusion on which is the most appropriate according to the evaluation metrics chosen. In this step, it is expected an in-depth analysis of the results that goes beyond simply reporting the results;
7. **Report – Paper Draft:** you must draft a manuscript reporting the entire process you have made during the project. Details on what and how this manuscript must include are given below;
8. **Presentation:** you must prepare a slide deck that will be presented to an assessment committee. The presentation details and date will be made available on Canvas/AVA.

In this project, you have **three** items to deliver:

- The source code;
- The manuscript;
- Slide deck.

Each of these items is further explained as follows.

Details on the source code

Throughout this project, you and your team will have to fulfill each step of the Python notebook template provided, including mainly:

- Statistical description of the dataset;
- Univariate analysis;
- Multivariate analysis;
- Effective data visualization;
- Data selection and preprocessing;
- Machine learning and tuning;
- Validation.

Detailed information about each topic can be found in the project template. A partial delivery of the report is expected.

Details on the manuscript (scientific report/paper)

In addition to implementing predictive models, your team's work must be detailed in the format of a scientific paper. This paper's manuscript must have **at least** the following information:

- 1) Definition of the dataset and task being tackled;
- 2) The experimentation protocol adopted, i.e., how the dataset was split into training, validation, and test sets, which validation metrics have been used, etc;
- 3) Which predictive models and hyper-parameters have been tested;
- 4) Which preprocessing steps have been used and why;
- 5) Results that have been obtained using validation data;
- 6) A discussion on such results.

Important details about the manuscript:

- The aforementioned items do not reflect the sections that the manuscript should contain;
- The manuscript must have between 4 and 6 pages (references included);
- For guidance on how to structure a scientific paper, please refer to the examples provided in Canvas;
- The manuscript must be prepared using IEEE's conference template. The template in both Word and LaTeX formats can be found here: <https://www.ieee.org/conferences/publishing/templates.html>;
- The references and citation style should be consistent with the template;
- The use of Generative Artificial Intelligence (AI) during manuscript drafting, e.g., chatGPT, is forbidden.

General considerations:

- **The delivery dates are provided in Canvas/AVA. Please check them before asking the turn-in dates for both the checkpoint and final deliveries;**
- All teams must have 6 students.
- Delivery dates must be respected by **ALL** students. **Delayed** deliveries will be **graded zero**;
- Submissions will be **DISQUALIFIED and graded 0 (zero)** if the manuscript does not reflect the source code;
- Submissions will be **DISQUALIFIED and graded 0 (zero)** if the source code does not run.